

Spec — CRM Custom Views (List, Kanban, Split)

App	TAVI CRM — https://crm.taviportal.com (tenant NDC-Staging)
Feature spec (PRD)	NDC-1868 — local copy <code>Test Cases/CRM/custom-views-spec.md</code>
Version	v3 — draft for review (2 Oct 2026). v2 = <code>spec-crm-custom-views-v2.md</code> ; see "Changes in v3" below
Gap ticket	NDC-1869 — 22 missing functions (Z5–Z27); each of those Z rows names its NDC-1869 item number. Z28–Z55 are the 11 new gaps from the live Zoho check (missing list Part B) and are not filed yet
Requirements	<code>docs/custom-views/custom-views-requirements-v3.md</code> — every requirement (<code>CVR-...</code>) points to scenarios here
Missing list	<code>docs/custom-views/custom-views-missing-list.md</code>
Test cases (v3)	<code>docs/custom-views/Custom Views - Spec v3 Test Cases.xlsx</code> (ids <code>CVS-<scenario id></code>)
Related	NDC-1836 (queues vs views), NDC-1317 (Kanban defects), NDC-1644 (Work Queue), NDC-1478 (Missing module views)
Existing test cases	<code>Test Cases/CRM/Custom Views.xlsx</code> (CV-001...CV-179), mapped in §Traceability
Reference product	Zoho CRM — List View, Kanban View, Module Views, Custom View API v8
Written	1 Oct 2026, from bundle <code>index-DfcbYBNN.js</code> + live API + UI (owner account); v2 re-checked against the Zoho list-view FAQ, list-view and NextGen list-view help pages, and the same build

Changes in v3

- Zoho CRM checked live** (2 Oct 2026, Leads module, admin account): view picker, Manage Custom View page, New Custom View form (operator lists per field type, sharing picker, lock), list view grid (header menu, column settings, row menu, selection bar, module menu), filter panel and Kanban settings dialog. Nothing was created in Zoho.
- New Group Z rows Z28–Z55** for the 11 new gaps (missing list Part B): saved filters, filter by related modules, Lock this View, owner criteria by role, Previous / Next N units, text operators on pick lists, Wrap Text, Reset Column Size, A–Z letter filter, bulk delete of views, more selection and module actions. They are not in NDC-1869 yet.
- Z7 corrected**: live Zoho has fiscal quarter and fiscal year operators only — no plain calendar-quarter operator. The build does have a tenant fiscal-year setting (Company settings → Fiscal year), so Z7 is no longer blocked by a missing setting.
- Z24 corrected**: "Set reminders" is not in live Zoho's selection menu and is removed from the expected result.
- J5 corrected**: Zoho's Sheet View article says 100 existing rows plus 200 new rows; the list-view FAQ says 999. Both are quoted.
- New D25**: Last Activity Time as a criteria field (NDC-1868 P1-5) — Zoho offers it (seen live); to be run on CRM.
- Zoho details confirmed live and added to the expected results of existing gaps: sharing picker kinds (Z8), header menu items (Z12), row menu items (Z20), System Defined Filters list (Z25), Manage page grouping (Z27).
- Traceability extended with the new requirements; a requirements document was added (see the table above).

Changes in v2

- Re-checked against Zoho docs** — [FAQs: List Views](#), [Managing List Views](#), [Managing List Views \(NextGen\)](#), [Managing Profile Permissions](#).
- New Group Z rows Z19–Z25** (column resize, row actions menu, activity/notes badges, Converted/Junk Leads views — later withdrawn, see item 12 — "Coming soon" actions, missing bulk actions, unavailable system-defined filters).

3. **F12 and J5 corrected:** Sheet View, Print View, Drafts, Mass Convert, Mass Email (and Approve, Deduplicate, Add to Campaigns, Create Client Script by code) show a "Coming soon" toast — confirmed on screen 1 Oct 2026. Only Delete Selected, Mass Delete, Mass Update, Export, Manage Tags, Import and Import History have handlers.
4. **New F16–F19** for what the re-check showed about the grid (header click sorts directly, no resize handle, rows have inline "Edit field" only, filter panel sections not available).
5. **D21 expected result fixed:** build and Zoho both read 1 or 2 and 3 left to right as ((1 or 2) and 3) (build help text: "Conditions are read from left to right").
6. Table rows that had an extra "steps" cell under a 4-column header (F10–F14, D23, G4, G8–G13, I11, K2–K5, K7, K9–K11) fixed: the steps now follow the scenario after a colon.
7. Traceability updated for the new rows.
8. Correction for NDC-1869 (applied 1 Oct 2026): Zoho **does** support column width resize ("You can edit and change the order or size of columns, but you can not merge them"), and Sheet/Print views are **not** available in CRM (Coming soon).
9. **Favourites withdrawn as a gap (Z1):** current Zoho CRM no longer has "Mark as favourite"; it was replaced by **Pin view** (the help pages above still describe the old favourites option). CRM has Pin view, so Z1 is not a gap; G5 now covers it and is the test for the PRD's favourite criteria (P0-1, P0-2; CV-001, CV-018, CV-019). NDC-1869 item 1 should be removed when the ticket is updated (source: user, 1 Oct 2026).
10. **Feature inventory added** (below): built / missing / not a gap / not yet confirmed, each pointing to its test ids.
11. **Picker groups and Manage custom views page** (user notes, 1 Oct 2026): each picker group should collapse/expand, and the Manage custom views page should group views by type (Created by me, Shared with me, Public views, Pinned views — Pinned only once a view is pinned). Checked on the live build the same day: group names are plain labels (no collapse), and the Manage page is one flat table with "All custom views" / "Public views" tabs. Added as expected behaviour in B1 and new B16, and as proposed gaps Z26 and Z27.
12. **Z2, Z3, Z4 and Z22 withdrawn** (user note, 1 Oct 2026): Other users' views for admins, setting a default view, the Recently Viewed standard view, and the Converted Leads / Junk Leads views are not gaps. Their Z rows are kept with strike-through so numbering stays stable, moved to "Not a gap" in the inventory, and removed from traceability. NDC-1869 items 2, 3 and 4 should be removed when the ticket is updated (Z22 was never filed).
13. **NDC-1869 updated to match v2** (1 Oct 2026): old items 1–4 removed, the new gaps added, the column-resize and Sheet/Print statements corrected, and the items renumbered 1–22. Each Z row now names its NDC-1869 item number. The NDC-1868 comment now says 22 missing functions.

Feature inventory

What the CRM build has today compared with current Zoho CRM, as found on 1 Oct 2026 (bundle `index-DfcbYBNN.js`, live API, UI, owner account). Each line points to the scenarios that test it.

Built in CRM

View picker and standard views

- View picker with search, tabs, "Show custom views as a dropdown/tabs", and the groups Recently closed, Pinned views, Created by me, Shared with me and Public views. The Pinned views group appears only once a view is pinned. **Expected but not built yet:** each group can be collapsed and expanded to hide or show its views — today the group names are plain labels (Z26) — B1–B6, Z26
- Manage custom views page: a single table (View Name, Shared To, Created By, Last Modified By, Layout, Actions) with search and two tabs, "All custom views" and "Public views". **Expected but not built yet:** views grouped by type — Created by me, Shared with me, Public views, and Pinned views (shown only once a view is pinned) (Z27) — B16, Z27
- Five standard views: All, My, Recently Created, Recently Modified, Today's. They cannot be renamed, cloned or deleted; their columns, sort and page size can be changed — B7–B10, A7, A12
- Last used view remembered per module, across opening a record and signing out — B11–B14
- Pin view (up to 15), current Zoho's replacement for favourites — G5, G6

Create and edit a custom view

- Name (required, up to 120 characters, unique per owner), criteria up to 25 rows, layout scope — C1–C8, C13
- AND/OR pattern with brackets, read left to right like Zoho — D20–D24
- Operators per field type: text, number, date (Today, Yesterday, Tomorrow, this/last/next week, month and year, Age in Days, Due in Days, last/next N days), checkbox, pick list, multi-select, lookup — D1–D16
- Owner = me, tags and review status as criteria — D17, D18, E12
- Lookup field criteria (up to 5 rows) — E8
- Related module criteria with "with" and "without" (up to 3 modules) — E1–E7
- Warning when a criteria field is deleted or hidden — E9–E11
- Columns (up to 30), sort, records per page — C9–C12
- Sharing: Only me, Everyone, Selected users (sharing needs "Manage Shared Views") — C14, C15, A2–A6
- Save-conflict check when two people edit the same view — C18

List view grid

- Manage Columns, plus a personal column change that only you see (Save to view / Use the view's columns) — F1–F4
- Click a header to sort, paging, record count, select the rows on a page — F5–F9, F16
- Inline "Edit field" in a row — F14
- Working actions: Delete Selected, Mass Delete, Mass Update, Export, Manage Tags, Import, Import History — F11–F13

Clone, pin and delete

- Clone (not for system views), delete with confirmation to the Recycle Bin, restore, Manage custom views page — G1–G4, G7–G13

Kanban

- Kanban settings: categorize by pick list, status or radio field, aggregate (number, currency, decimal, percent, formula), mono or multi colour header, up to 10 card fields, sharing, several boards per module, Manage Kanban views — H1–H13
- Board: Unaccounted column, counts and totals from the server, Load more (100 at a time), collapse and expand, create from a column, layout and pipeline pickers, stage probability, 75-column limit notice, refresh every 30 seconds — H14, I1–I5, I14–I20
- Drag-and-drop saves through the normal record update, follows Blueprint and required fields, and can be done with the keyboard — I6–I13

Other

- Arabic translations and RTL — L1–L3, D24
- Split view — J1–J4

CRM extras that Zoho does not have

- Full create / edit / delete / clone / pin API for views — \$2.1, K2–K5
- Save-conflict check on views — C18
- Deleted views go to the Recycle Bin and can be restored — G7, G8
- "In the last / next N hours" and "Before now / After now" operators — D11, D12
- Keyboard way to move Kanban cards — I13

Missing in CRM

Z1, Z2, Z3, Z4 and Z22 are withdrawn (see "Not a gap"). Z5–Z27 are in NDC-1869 (22 items); the Group Z table gives each one's item number. Z28–Z55 are new in v3 and not filed yet.

- **View picker:** Z26 collapse and expand each picker group · Z27 Manage custom views page grouped by type
- **Criteria:** Z5 number between / not between · Z6 Starting tomorrow / Till yesterday · Z7 quarter and fiscal-year operators
- **Sharing and permissions:** Z8 share with roles and groups · Z9 permission to create custom views · Z10 permission to manage Kanban views

- **List view grid:** Z11 pin (freeze) a column · Z12 header menu with Filter by / Hide · Z13 select all records in the view · Z19 resize column widths · Z20 row actions menu · Z21 activity and notes badges
- **Kanban:** Z14 sort cards · Z15 reorder Kanban views, creator on hover · Z16 aggregate by rollup field
- **Deleting a view:** Z17 warn before deleting a view a Work Queue queue uses (today the queue shows "source view unavailable" afterwards)
- **Actions and filters:** Z23 actions that only show "Coming soon" (Mass Convert, Mass Email, Drafts, Approve, Deduplicate, Add to Campaigns, Create Client Script, Sheet View, Print View) · Z24 Run macro, Create task, Set reminders, Change owner for many records, Bulk mail merge · Z25 System-defined and Website Activity filters
- **Other view types:** Z18 Chart, Timeline, Grid, Map, Canvas (PRD non-goal)

New in v3 — seen in live Zoho on 2 Oct 2026, **not in NDC-1869 yet** (missing list Part B):

- **Filters:** Z28–Z31 Save filter · Z32–Z33 Filter By Related Modules
- **Create / edit a view:** Z34–Z37 Lock this View · Z38–Z39 owner "belongs to Role" · Z40–Z42 date "Previous / Next" N days, weeks, months, years · Z43–Z44 text operators on pick lists
- **List view grid:** Z45–Z46 Wrap Text view mode · Z47 Reset Column Size · Z48–Z49 A–Z letter filter
- **Manage custom views:** Z50–Z52 delete several views at once
- **Actions:** Z53 Cadences, Print Mailing Labels, Print Using Canvas · Z54 Export Selected Records · Z55 Assignment Rules, Mass Transfer

Not a gap

- Favourites: current Zoho replaced them with Pin view, which CRM has — Z1 (withdrawn), G5
- "Other users' views" group for admins — Z2 (withdrawn; confirmed not a gap by the user, 1 Oct 2026)
- Setting a default view per module — Z3 (withdrawn; confirmed not a gap by the user). CRM opens the last used view, or All — B11–B14
- "Recently Viewed" standard view — Z4 (withdrawn; confirmed not a gap by the user). CRM's standard views are All, My, Recently Created, Recently Modified, Today's — B7
- "Converted Leads" and "Junk Leads" standard views — Z22 (withdrawn; confirmed not a gap by the user)
- 1 or 2 and 3 without brackets: CRM reads it left to right like Zoho — D21
- Following records from the list view: Zoho cannot do it either
- Merging columns: Zoho cannot do it either
- Work Queue queues and module views kept apart: already filed as NDC-1836
- No API to create or delete views: Zoho has none either; CRM has one (extra)

Not yet confirmed

- Everything marked 🛡️ (who can share, edit standard view columns, edit or delete other users' views, drag cards): no working lower-profile account — §1.4
- Subform criteria (PRD P1-2): no module with a subform on NDC-Staging — §1.4
- Last Activity Time as a criteria field (PRD P1-5): the Leads grid has a "Last Activity" column, but it hasn't been checked as a criteria field yet — Traceability P1-5
- Last Activity Time as a criteria field: now has its own scenario, D25
- Date "isn't" and "is not empty" (Zoho has both): not in the build's date operator list in §2.3 — check in D1
- Zoho Kanban sort, reorder and rollup (Z14–Z16) and "Select all records in this view" (Z13): not seen live (needs a Kanban view created in Zoho, and more records than one page); they rest on Zoho's help pages

Tags used on every scenario: [func] functional · [edge] edge/boundary · [neg] negative · [data] data-class sweep · [ui] rendering/i18n/RTL · [ux] usability/a11y · [perf] performance · [sec] security/permissions/API · [int] cross-module · [compat] compatibility · [e2e] journey · 🛡️ = must run as a second, lower-profile account.

0 How to run

1. **Order:** §1 preconditions → A → B → C → D → E → F → G → H → I → J → K → L → E2E → Z. A fresh baseline (§1.3) is taken before A and diffed after the last group.
2. **Serial only:** C, G, H, I and every scenario marked [serial] write views or records. Run them one at a time — views are shared tenant data and another operator works in this tenant (ZZBUG/ZZRVW fixtures).
3. **Parallel allowed:** A, B, D (read parts), K-read, L-ui, all API GET scenarios.
4. **Exclusive:** [perf] scenarios run alone, with nothing else signed in to the account.
5. [data] rows record: field, operator, input value (exact bytes for odd input), what was saved (GET the view), what the grid returned (count + 3 sample ids), and pass/fail per row.
6. **Sessions:** the account is capped at 5 concurrent sessions (429 max_sessions_reached). Every script signs in for itself and signs out through the UI (profile → **Logout**); reload the page before signing out, because open panels cover the profile button. The access token lives ~5 minutes — refresh with POST /iam/auth/refresh .
7. **Verdicts:** Pass / Fail / Blocked / Inconclusive. A scenario blocked by an environment precondition (§1.4) is **Inconclusive, not Fail**.

1 Preconditions

1.1 Accounts

Role	Account	Status
Tenant Owner (admin)	CRM_USER_EMAIL in Automation/CRM-TAVI-Automation/.env (ndc-staging-owner)	Works
Second user, lower profile (Manager or User) 	needed	Blocker — CRM_USER_ALT1 signs in but the SPA stays on "Signing in..."
User of another tenant	throwaway accounts in Test Data/AnywareStagingTempAccounts.txt	Works (different tenant) — used for K-isolation only

Never print passwords or tokens in evidence. Report token **lengths** only.

1.2 Test data

- Prefix every created view, Kanban view, split view and record with QA CV (e.g. QA CV leads-contains).
- Modules: Leads 841043e1-c078-41f9-8d6e-dfd61bee45c3 , Deals f42cbafd-ab4e-4a6b-a40c-02c1d0ecb430 , Contacts 6f553963-01d6-4b68-9fb3-143d23044c41 , Tasks a9d87b62-... , Subscriptions 6d053261-... .
- Seed records (create via UI/API in C-setup, [serial]): 12 Leads named QA CV L01...L12 covering every Lead Status value, one with empty status, one with Arabic name, one with whitespace-only Comment, one with 255-char Company; 8 Deals QA CV D01...D08 across stages with Amount 0, 9999.99, 10000, 10000.01, empty, negative (if allowed), 15000, 1e9.
- Required fields: Leads require single_line_3_9 on every PATCH (PATCH replaces the whole data array).

1.3 Baseline (restore point)

Before group A, save:

- GET /modules/{m}/views?kind=list|kanban|queue for Leads, Deals, Contacts, Tasks, Subscriptions → baseline/views-<module>-<kind>.json (ids, names, row_version , settings).
- GET /modules/{m}/list-preferences for Leads and Deals → baseline/prefs-<module>.json .

- `GET /workqueue/queues` → `baseline/queues.json` . Revert = delete every `QA CV` object, restore preferences, and prove it with the final diff (\$Reporting).

1.4 Environment blockers

Blocker	Scenarios affected	Action
No working lower-profile account	every  scenario	Inconclusive until provided (requested in NDC-1869)
No module with a subform	Z-subform, E-subform	Inconclusive
Fiscal year	Z7	Not a blocker any more: the build has Company settings → Fiscal year. Record its start month before running Z7
Only Deals has a pipeline + blueprint-governed stage (<code>group_by_blueprint_governed:true</code>)	I-blueprint	Use Deals

1.5 Things that cannot be undone cleanly

- Deleted views go to the **Recycle Bin**, not away — purge them from Settings → Recycle Bin at cleanup, or record their ids in the report.
- Kanban drags are **record edits**: they fire workflow rules, audit log entries and Blueprint transitions. Only drag `QA CV` records, and check there are no active workflow rules on the target field that send email (`GET /workflow/rules`).
- Pinning a view/queue is per user and capped at 15 — unpin everything you pinned.

2 Surface and contract

2.1 Admin API (all under /api/v1, headers Authorization: Bearer , x-tenant-id , x-app-key: crm)

Method	Path	Notes
GET	/modules/{m}/views?kind=list kanban queue[&layout_id=]	{views:[...]} ; default kind in the client is kanban
GET	/modules/{m}/views/{v}	one view; 404 NOT_FOUND for unknown id
POST	/modules/{m}/views	create
PATCH	/modules/{m}/views/{v}	update; row_version → VERSION_CONFLICT
DELETE	/modules/{m}/views/{v}	to Recycle Bin
POST	/modules/{m}/views/{v}/clone	
PUT / DELETE	/modules/{m}/views/{v}/pin	pin / unpin (limit 15)
GET	/modules/{m}/related-modules	modules with a lookup to {m} and their link_fields
GET	/modules/{m}/records/board-summary? group_by=&aggregate=&filters=&q=&layout_id=&view_id=	per-column count , sum , unaccounted , truncated , group_by_blueprint_governed
GET / PUT	/modules/{m}/list-preferences	per-user: lastListViewId , openListViewIds , recentlyClosedListViewIds (≤10), column overrides (≤50), pageSize (25), lastView list/kanban/split, lastKanbanPipelineId
GET	/workqueue/queues	Work Queue aggregation of kind=queue views
—	/views , /custom_views , /settings/custom_views	404 (no Zoho-style endpoint)

View object: id, module_id, layout_id, kind, name, owner_user_id, visibility (private|everyone|selected), shared_user_ids[], settings, sort_order, row_version, created_*, last_modified_*, can_edit, can_delete, is_system, system_key, pinned, pin_order, category (created_by_me|shared_with_me|public_views) .

settings for list: {sort:{field,dir}|null, columns:[], criteria:{pattern, conditions:[{field,operator,value}]}|null, page_size|null, related_criteria:[{module_id, link_field, mode:with|without, criteria?}]} . settings for kanban: {categorize_field, aggregate_field|null, header_style:mono|multi, card_fields:[]} . settings for queue: list settings + source: criteria|view|date (+ view_id) .

2.2 Field register (limits found in the client bundle — server must enforce the same)

Item	Kind	Limit
View name	text	required, ≤120 chars, unique per owner per module (VIEW_NAME_TAKEN)
Criteria rows	list	≤25
Lookup-path rows	list	≤5
Related modules	list	≤3 blocks, ≤3 rows each
Pattern	text	≤200 chars, digits/and/or/brackets, nesting ≤3
Columns	list	30 max, tip shown above 10
Page size	int	10–200
last_n_days / next_n_days / age_in_days / due_in_days	int	0/1–3650
last_n_hours / next_n_hours	int	1–720
Pinned views	count	≤15
Kanban card fields	list	≤10 (default 3), record name always first
Kanban columns	count	≤75, notice "Only the first {{max}} values are shown as columns."
Kanban column page	int	100 records per "Load more"
Board summary refresh	timer	30 s
Shared users	list	users only (no roles/groups)

2.3 Operators per field type (build)

- Text: equals, not_equal_to, contains, does_not_contain, starts_with, ends_with, is_empty, is_not_empty.
- Number/currency/decimal/percent: equals, not_equal_to, greater_than, less_than, ≥, ≤, is_empty, is_not_empty. (no between — Z5)
- Date: is, is_before, is_after, is_between, is_not_between, today, yesterday, tomorrow, this/last/next week, this/last/next month, this/last/next year, last_n_days, next_n_days, age_in_days, due_in_days, before_now, after_now, empty. (no starting tomorrow / till yesterday — Z6; no fiscal — Z7; no previous / next N weeks, months, years — Z40; Zoho also has "isn't" and "is not empty")
- Datetime: date set + last_n_hours, next_n_hours.
- Checkbox: is_true, is_false.
- Picklist/status/department/team: equals, not_equal_to, contains_any_of, contains_none_of, empty. (no contains / starts with / ends with — Z43)
- Multi-select: contains_any_of, contains_all_of, contains_none_of, empty.
- Lookup/pipeline: equals, not_equal_to, empty. (no "belongs to Role" on owner fields — Z38)
- Synthetic fields: owner (incl. \$me), source, tags, review_status (permission-gated).

2.4 Page states and messages (en)

Picker: "Tick the views to show as tabs, or click a name to open it." · groups Recently closed / Pinned views / Created by me / Shared with me / Public views · footer "Show custom views as a dropdown|tabs", "New custom view", "Manage custom views", "** System-defined custom view" · empty "No views match." · group names are plain labels (not collapsible); Pinned views shown only when a view is pinned. Manage custom views (/modules/{m}/views/manage?layout=...): "Manage {{module}} custom views", "Create custom view", search "Search by name", tabs "All custom views" / "Public views", columns View Name · Shared To ·

Created By · Last Modified By · Layout · Actions, legend "*" System-defined custom view", empty "No custom views match.", error "The custom views could not be loaded." Tab menu: Edit, Edit columns, Pin view / Unpin view, Clone ("You cannot clone a system-defined view."), Close view, Delete view. Editor (/modules/{m}/views/new?layout=... , /views/{v}/edit): "New custom view for {{module}}", "Applies to the "{{layout}}" layout — its fields and its records.", "Enter a name for the view.", "A view name can be at most {{max}} characters.", "You already have a view with this name.", "Fix the criteria pattern: every row number must appear once, and brackets must match.", "At most {{max}} criteria rows can use fields of lookup modules.", "A view can use at most {{max}} related modules.", "No columns chosen: everyone sees their usual columns.", "Tip: more than {{max}} columns makes the table hard to read.", "You need the Manage Shared Views permission to share a view with other people.", "Choose at least one user, or share with everyone instead.", "System-defined views keep their name and criteria. You can change their columns, sort and page size.", "Some fields in this view's criteria are deleted or hidden from you: {{fields}}...", "Someone else saved this view after you opened it. Reload...", "View not found — This view was deleted, or it is not shared with you.", "You cannot edit this view — Only its owner, or someone who can manage shared views, can change it." Delete: "Delete custom view? Are you sure you want to delete the "{{name}}" custom view? It moves to the Recycle Bin; its records are not affected." Columns override: "You changed the columns for this view. Only you see this change." · "Save to view" · "Use the view's columns". Kanban: "Add a pick list, status or radio-button field to this module to use the Kanban view", "Create a Kanban view to use this board", "No Kanban views yet", "Unaccounted", "This value is no longer an option of the field", "Probability {{value}}%", "Cannot update as the record is under the Blueprint", "Choose the transition to {{column}}", "{{field}} is required, so a record cannot be left without a value.", "Could not move the record.", "The board is still loading — try again in a moment.", "Only the first {{max}} values are shown as columns.", delete ""{{name}}"" will move to the Recycle Bin. The records it shows are not affected." Unavailable: "Custom views are not available — Custom views are turned off for this workspace."

2.5 Data classes (D1–D20; _COMMON.md is not on this machine, so they are listed here)

D1 empty · D2 whitespace only · D3 single char · D4 max length · D5 max+1 · D6 leading/trailing spaces · D7 case variants · D8 Arabic · D9 mixed Arabic/Latin + RTL marks · D10 emoji/astral · D11 HTML/script () · D12 SQL-ish (' OR 1=1 --) · D13 path/URL chars (../ , %2e) · D14 zero-width/control chars · D15 numeric edge (0, negative, decimal, 1e9, 15,000 , Arabic-Indic ١٥٠٠٠) · D16 date edge (29 Feb, DST, 23:59 Cairo vs UTC) · D17 type confusion (number as string, array for scalar, object, null) · D18 unknown field / operator id · D19 duplicate (same name / same row twice) · D20 very long list (limit+1 items).

Group A — Access and permissions

ID	Scenario	Steps	Expected	Tags
A1	Owner sees full view UI	Open Leads as owner	Tabs, picker, New custom view, Manage custom views, Kanban/Split switchers present	[func]
A2	Lower profile can create a private view	 user without Manage Shared Views → New custom view → save QA CV private	Saved; Everyone/Selected disabled with "You need the Manage Shared Views permission..."	[sec] 
A3	Sharing gate in API	 same user: POST view with visibility:"everyone"	403/422; no public view created	[sec] 
A4	Recipient cannot edit/delete shared view	 owner shares QA CV shared with user → user opens Edit URL and calls PATCH/DELETE	UI shows "You cannot edit this view"; API refuses; can_edit=false , can_delete=false	[sec] 
A5	Private view by id	 user GETs owner's private view id, opens /modules/{m}/views/{id}/edit	404 / "View not found"; no settings leak	[sec] 
A6	Selected-users outsider	 view shared with user X; user Y lists views and GETs id	Not listed; GET refused	[sec] 
A7	Standard view columns by non-admin	 user opens All Leads → Edit columns → Save to view	Only a personal override ("Only you see this change"); system view row_version unchanged	[sec] 
A8	Kanban drag without edit rights	 user without metadata.records.manage opens board	Cards not draggable; PATCH refused	[sec] 
A9	Other tenant	Sign in as a throwaway tenant user; GET a NDC-Staging view id with own token	404; nothing returned	[sec]
A10	No token / expired token	Call GET views without bearer, then with a 6-min-old token	401 TOKEN_MISSING / expired; no data	[sec]
A11	Missing tenant headers	Call with bearer but no x-tenant-id	Same result as with header, or 400 — record which; never another tenant's data	[sec]
A12	Direct URL to editor of system view	/modules/{m}/views/{allLeadsId}/edit	Opens with "System-defined views keep their name and criteria..."; name and criteria read-only	[func]
A13	Feature turned off message	(if a workspace with custom views disabled exists) open module	"Custom views are not available"; else mark Inconclusive	[func]

Group B — Picker, tabs, standard views, persistence

ID	Scenario	Steps	Expected	Tags
B1	Groups	Create one view of each category (mine, shared to me  , public) → open picker	Headings in order: Recently closed, Pinned views, Created by me, Shared with me, Public views; system views marked * ; Pinned views appears only once a view is pinned and disappears when the last one is unpinned; empty groups are not shown; each group can be collapsed and expanded to hide/show its views (today: plain labels, no collapse — Z26)	[func]
B2	Search	Type part of a name, a name in another case, Arabic name, a string with no match	Filters across groups; case-insensitive; "No views match."	[func] [data]
B3	Tabs	Tick/untick views to show as tabs; reload	Tabs persist (<code>openListViewIds</code>)	[func]
B4	Pinned tab cannot be hidden	Pin a view, try to untick its tab	Blocked with "Pinned views always show as tabs..."	[edge]
B5	Close view → Recently closed	Close a tab	Appears under Recently closed; max 10 kept (close 11)	[edge]
B6	Dropdown mode	"Show custom views as a dropdown" then back to tabs	Layout switches and persists	[ux]
B7	Standard views present	Each of Leads, Deals, Contacts, Tasks, Subscriptions	All, My, Recently Created, Recently Modified, Today's <code>{{module}}</code> ; criteria behave as named	[func]
B8	"My" view	Owner vs  user	Shows only records owned by the viewer	[func]
B9	"Today's" view at day boundary	Record created 23:50 and 00:10 Cairo time	Uses tenant timezone (Cairo), not UTC	[edge]
B10	Standard view actions	Open menu on All Leads	Edit columns, Pin, Close only; Clone disabled with tooltip; no Delete/Rename	[func]
B11	Last view survives navigation	Select QA CV x , sort, page 2, open a record, Back	Same view, sort, page (NDC-1317)	[func]
B12	Last view survives sign-out	Select view, sign out/in	Restored from <code>lastListViewId</code>	[func]
B13	Per module	Change view on Leads; open Deals	Deals unchanged	[func]
B14	?view= URL	Open <code>/modules/{m}?view={id}</code> for own, shared, deleted, foreign ids	Own/shared open; deleted/foreign fall back to All with no crash	[edge] [sec]
B15	Two tabs preferences	Tab 1 opens view A, tab 2 view B, close tabs in each	Record final <code>list-preferences</code> ; lost update = possible bug (NDC-1869 note)	[edge]

ID	Scenario	Steps	Expected	Tags
B16	Manage custom views page	Picker → Manage custom views (/modules/{m}/views/manage); search; switch tabs; pin and unpin a view, reload	Lists every view the user can see with View Name, Shared To, Created By, Last Modified By, Layout, Actions; search filters by name; views grouped by type — Created by me, Shared with me, Public views, and Pinned views only while at least one view is pinned (today: one flat table with "All custom views" / "Public views" tabs — Z27)	[func]

Group C — Create and edit a list view [serial]

ID	Scenario	Steps	Expected	Tags
C1	No criteria	Name only, Save	Created by me; returns every record the user can access; count equals All	[func]
C2	Name D-sweep	Name = D1, D2, D3, D4(120), D5(121), D6, D7 (vs existing), D8, D9, D10, D11, D14	D1/D2 → "Enter a name for the view."; D5 → "at most 120"; D6 trimmed (record); D7 → document rule; D11 rendered as text everywhere (picker, tab, manage page, delete dialog)	[data] [sec]
C3	Duplicate name same owner	Save QA CV dup twice	"You already have a view with this name." (VIEW_NAME_TAKEN)	[neg]
C4	Same name different owner	 user saves QA CV dup	Allowed	[edge] 
C5	Same name across kinds	Create list QA CV k and queue QA CV k	Record whether allowed (kinds are separate rows)	[edge]
C6	Incomplete row	Field + operator, no value → Save	Row flagged; save blocked	[neg]
C7	Value-less operator	is_empty / today / is_true	Saves without a value	[func]
C8	25 rows	Add 25 rows; try 26th	Add disabled at 25; API POST with 26 rejected	[edge] [sec]
C9	Columns	Choose 8 columns, reorder, Save	Grid shows exactly those, in order	[func]
C10	No columns	Save with none chosen	"everyone sees their usual columns" behaviour	[func]
C11	11 and 30 columns; 31 via API		Tip above 10; 30 allowed; 31 rejected server-side	[edge]
C12	Sort and page size	Sort by Created desc, page 10 / 200 / "Each person's usual"	Applied; 9 and 201 via API rejected	[edge]
C13	Layout scope	Create on Standard layout; switch layout	View applies only to that layout's fields/records ("Applies to the ... layout")	[func]
C14	Sharing — Selected with none	Selected users, no user chosen	"Choose at least one user, or share with everyone instead."	[neg]
C15	Edit propagates	Owner edits criteria of a shared view	 recipient's results change on next load	[func] 
C16	Prefilled from filters	Apply Smart Filter on list, then New custom view	"The criteria start with the filters applied on the records page." and rows prefilled	[func]
C17	Cancel	Fill form, Cancel / Back to records	Nothing created (GET list unchanged)	[func]

ID	Scenario	Steps	Expected	Tags
C18	Version conflict	Open edit in two tabs, save tab 1, save tab 2	Tab 2 shows "Someone else saved this view..." + Reload; no silent overwrite	[edge]
C19	Double submit	Double-click Save / two POSTs in parallel	One view only	[edge]
C20	Partial PATCH	PATCH with only {name}	Other settings preserved (record)	[edge]
C21	Read-only keys	PATCH is_system:true , owner_user_id:<other> , category , can_edit , row_version:999	Ignored or rejected; GET shows unchanged	[sec]
C22	Type confusion	POST with D17 in name, columns, criteria, page_size	400/422, no 500, no partial row	[sec] [data]
C23	Session expiry	Leave editor open > 5 min, Save	Token refresh then save, or clear sign-in prompt; no data loss without warning	[edge]
C24	Audit	After C1/C15/G-delete check Settings → Audit Log	Create/edit/share/clone/delete recorded with actor and time (PRD NFR)	[int]

Group D — Criteria operators and pattern

Run each [data] row via the editor and assert the grid count against an independent API filter (GET /modules/{m}/records?filters=...) with the same condition.

ID	Scenario	Expected	Tags
D1	Operator list per field type (§2.3)	Exactly the listed operators; none extra/missing	[func]
D2	Text: equals/not_equal/contains/does_not_contain/starts_with/ends_with with D3, D6, D7, D8, D10, D12, D14	Correct split; case rule same as module filter; D12 treated as literal	[data] [sec]
D3	Text is_empty / is_not_empty with null, "", whitespace-only	Rule documented and identical to module filter	[edge]
D4	Number: =, ≠, >, <, ≥, ≤ at 9999.99 / 10000 / 10000.01	Boundary exact	[edge]
D5	Number value D15: 15,000 , ١٥٠٠٠ , 1e9 , -1 , abc	Never stored as 15; normalised or rejected with message	[data]
D6	Number empty vs 0	Empty ≠ 0	[edge]
D7	Date is/before/after/between/not between incl. same-day bounds	Bounds rule stated and consistent	[edge]
D8	today/yesterday/tomorrow at 23:50 and 00:10 Cairo	Tenant timezone	[edge]
D9	this/last/next week, month, year	Calendar boundaries correct (week start documented)	[func]
D10	last_n_days / next_n_days / age_in_days / due_in_days with 0, 1, 3650, 3651, -1, 1.5	0/1–3650 accepted, others rejected	[data]
D11	Datetime last_n_hours / next_n_hours 1, 720, 721	721 rejected	[edge]
D12	before_now / after_now	Correct split at now	[func]
D13	Checkbox is_true / is_false (unset value?)	Unset treated as false or excluded — record	[edge]
D14	Picklist equals / not_equal / contains_any_of / contains_none_of incl. a value later removed from the picklist	Stale value handled, no crash	[edge]
D15	Multi-select any_of / all_of / none_of	Correct set logic	[func]
D16	Lookup equals / empty	Correct	[func]
D17	Owner = \$me	Each user sees own records 	[func] 
D18	Tags criteria	Records with tag only	[func]
D19	Unknown field / operator via API (D18)	400/422	[sec]
D20	Pattern: default AND for 3 rows	Results equal 1 and 2 and 3	[func]
D21	Pattern 1 or 2 and 3 (no brackets)	Read left to right as ((1 or 2) and 3) , same as Zoho; the editor help says "Conditions are read from left to right"; results match	[edge]

ID	Scenario	Expected	Tags
		that grouping, and the saved pattern reopens fully bracketed (PRD Q1)	
D22	Pattern errors: unbalanced, () , (and) , 1 and , 1 2 , 1 and 5 (row missing), unused row, row used twice, word xor , AND upper-case, 201 chars, depth 4	"Fix the criteria pattern..." (or specific message); save blocked; API rejects too	[neg] [data]
D23	Pattern round trip: save (1 or 2) and 3 , reopen	Same meaning; record canonical form	[func]
D24	Pattern in RTL	Arabic UI: row numbers, brackets and and/or read correctly	[ui]
D25	Last Activity Time as a criteria field: Age in Days = 7 on a lead with a note today and a lead with no activity	Field offered with the date-time operators; only the lead with recent activity is returned; adding a note brings the other lead in (NDC-1868 P1-5; Zoho offers the field)	[func]

Group E — Related modules, lookup paths, hidden fields

ID	Scenario	Expected	Tags
E1	Related: Contacts <i>with</i> Deals Stage = Negotiation	Contacts having ≥ 1 such deal	[func]
E2	Related: <i>without</i> Deals	Contacts with no deals	[func]
E3	Without + record the viewer cannot see 	Rule per PRD Q14 — record; leak of hidden existence = finding	[sec] 
E4	"Any link field" vs a specific link field (Deals has many lookups: Connected To, Lookup 2...)	Specific link narrows correctly	[func]
E5	3 related blocks; 4th	"A view can use at most 3 related modules." ; API refuses 4	[edge]
E6	3 rows in a block; 4th	Blocked	[edge]
E7	Module with no lookup to it	"No module has a lookup to {{module}} yet."	[func]
E8	Lookup path rows (e.g. Account Name › Industry) ≤ 5 ; 6th	"At most 5 criteria rows can use fields of lookup modules."	[edge]
E9	Criteria field later deleted	Banner "Some fields in this view's criteria are deleted or hidden..."; view still runs	[edge]
E10	Criteria field hidden for viewer 	Masked, read-only; view runs (PRD Q13 — record inference)	[sec] 
E11	Column of a field hidden for viewer 	Column absent and value absent from API response	[sec] 
E12	review_status synthetic field without review permission 	Not offered	[sec] 

Group F — Columns, sort, page size, selection in the grid

ID	Scenario	Expected	Tags
F1	Manage Columns: show/hide, required locked, reset	Works; record-name column cannot be hidden	[func]
F2	Personal override	Change columns on a shared view → "Only you see this change" → other user unchanged 	[func]
F3	Save to view / Use the view's columns	Owner saves override into the view; reset restores	[func]
F4	Override limit 50 views	Override 51 views (API) → oldest dropped or rejected — record	[edge]
F5	Header sort asc/desc on text, number, date, empty values	Correct order; empties placement consistent	[func]
F6	Toolbar sort control	Same results as header sort; label "Sort: field dir"	[func]
F7	Page size 10/25/50/100/200 and paging	Counts and page indicator correct; last page partial	[func]
F8	Count matches rows	Header "N records" = rows across pages = API total	[func]
F9	Select page	Header checkbox selects visible page; bar shows count	[func]
F10	Selection cleared on view change: select rows, then switch view	Selection cleared	[edge]
F11	Bulk action uses selection only: Mass Update on 2 selected QA CV leads	Only those 2 change	[int] [serial]
F12	Module actions menu: open every item (with and without a selection)	Working: Delete Selected, Mass Delete, Mass Update, Export, Manage Tags, Import, Import History open their screens. "Coming soon" toast: Mass Convert, Drafts, Mass Email, Approve Leads, Deduplicate Leads, Add to Campaigns, Create Client Script, Sheet View, Print View (see Z23). Items needing a selection are disabled with "Select at least one record" when nothing is selected	[int]
F13	Export from a view: Export Leads while QA CV view active	Record whether export respects the view criteria/columns	[int]
F14	Inline edit in grid: "Edit field" on a QA CV cell	Saves through record rules (validation, required)	[int] [serial]
F15	Empty view: criteria matching nothing	Empty state reads as empty, not error	[ux]
F16	Header click: click a column header (e.g. Lead Status)	Sorts directly (URL gets ?sort=lead_status:asc), click again → desc; no header menu today (see Z12)	[func]

ID	Scenario	Expected	Tags
F17	Column width: drag a header border	No resize today (see Z19); after build: width kept per user per view, shared definition unchanged	[func]
F18	Row hover: hover a list row	Today only inline "Edit field" buttons; no row actions menu, no activity/notes badge (see Z20, Z21)	[func]
F19	Filter panel sections: open Filter	Fields section works; "System Defined Filters" and "Website Activity" show "These filters aren't available yet — coming in a future update." (see Z25)	[func]

Group G — Clone, pin, delete, Work Queue dependency [serial]

ID	Scenario	Expected	Tags
G1	Clone own view	New private copy, all settings equal, original untouched	[func]
G2	Clone shared view	Copy owned by cloner, private	[func]
G3	Clone system view	Disabled "You cannot clone a system-defined view."; API refuses	[neg]
G4	Clone name collision: clone twice	Unique name generated or prompt — no VIEW_NAME_TAKEN 500	[edge]
G5	Pin / unpin (current Zoho's replacement for Favourites)	Pinned view appears first under "Pinned views", ordered by pin_order , tab always shown; unpin removes it from the group; works for own, shared and system views; per user (another user's picker unchanged	[func]
G6	Pin 16th	Refused with limit message	[edge]
G7	Delete confirmation	Names view; Cancel keeps it; Delete → Recycle Bin	[func]
G8	Restore from Recycle Bin: restore deleted QA CV view	View back with same id/settings — record	[int]
G9	Delete system view via API: DELETE All Leads id	Refused (can_delete=false)	[sec]
G10	Delete view used by a queue: create queue with source = view QA CV src ; delete the view	Expected (PRD P0-7): warning listing the queue. Actual today: deleted silently; queue shows "source view unavailable" (Z17)	[int]
G11	Deleted last-used view: user had it selected → reopen module	Falls back to default with no error	[edge]
G12	Admin deletes another user's view : owner deletes user's public view	Allowed; user's picker updates	[func]
G13	Unshare: owner removes user from Selected	View leaves user's picker; if selected, fallback	[func]

Group H — Kanban view settings [serial]

ID	Scenario	Expected	Tags
H1	No Kanban views yet	"No Kanban views yet" + Create view	[func]
H2	Create on Leads by Lead Status	Board appears; columns in picklist order + Unaccounted	[func]
H3	Categorize-by choices	Only single-value picklist, status, radio fields	[func]
H4	Module without such field	"Add a pick list, status or radio-button field..." / "This module has no field a board can group by."	[neg]
H5	Name D-sweep (D1, D4, D5, D11)	"Give the view a name." ; limits; text escaped	[data] [sec]
H6	Aggregate choices	number, currency, decimal, percent, long int, formula; rollup excluded ("Rollup summaries cannot be totalled.")	[func]
H7	Card fields max 10	11th disabled; order up/down; record name first	[edge]
H8	Header mono / multi	Rendered	[ui]
H9	Sharing like list views; no Manage Shared Views → only me 	Message "You can create views for yourself..."	[sec] 
H10	Several Kanban views per module, different fields	Chooser switches; "Shared by {{name}}" shown for others'	[func]
H11	Edit categorize field	Board rebuilds	[func]
H12	Delete Kanban view	Confirmation → Recycle Bin; records unaffected	[func]
H13	Manage Kanban views page	Lists views; Create Kanban View	[func]
H14	76+ picklist values	Notice "Only the first 75 values are shown as columns."; record where beyond-cap records go (PRD P1-7 wants "Other values")	[edge]

Group I — Kanban board and drag-and-drop

ID	Scenario	Expected	Tags
I1	Counts and sums are server-side	Header count/sum = board-summary before scrolling	[func]
I2	Load more	Column with >100 records: Load more until all; final count = header	[func]
I3	Unaccounted column	Empty-value records appear there	[func]
I4	Stale value	Record with a value removed from picklist → "This value is no longer an option of the field"	[edge]
I5	Deals by Stage: pipeline picker + probability	Picker shown; "Probability N%" per stage	[func]
I6	Drag QA CV lead between statuses [serial]	GET record shows new value; both column counts/sums update; audit + workflow as a normal edit	[func] [int]
I7	Drag to Unaccounted when field required	"{{field}} is required..."; card returns	[neg]
I8	Drag on blueprint-governed Deals stage	"Choose the transition to {{column}}" or "Cannot update as the record is under the Blueprint"; no bypass	[sec] [int]
I9	Drag to Closed Won/Lost	Closing dialog; mandatory fields enforced	[func]
I10	Drag while loading	"The board is still loading — try again in a moment."	[edge]
I11	Concurrent move: two sessions move same card	Second gets conflict/refresh; no lost update	[edge]
I12	Drop on same column	No PATCH	[edge]
I13	Keyboard move	Move card via keyboard / menu, same rules	[ux]
I14	Collapse/expand one and all; drop on collapsed	Works	[func]
I15	Create from column footer	New record form pre-set to column value + layout/pipeline	[func]
I16	Select all loaded in column; bulk bar	Count correct; "Select all loaded"	[func]
I17	Board respects record access 	User sees only own-accessible deals; sums only over them	[sec] 
I18	30 s refresh	Change a record in another tab → header updates within 30 s	[func]
I19	Back-navigation	Open card, Back → same Kanban view, layout, pipeline, scroll (NDC-1317)	[func]
I20	Board with filter / search	filters / q applied to summary and cards consistently	[func]

Group J — Split view, Sheet and Print entry points

ID	Scenario	Expected	Tags
J1	Split view empty state	"No split views yet" + Create view	[func]
J2	Split by field / owner / criteria	Sections per value; counts; collapse; per-section columns	[func]
J3	Split limits	"too many conditions" message; max values	[edge]
J4	Records per split; create in section	Works	[func]
J5	Sheet View / Print View from module actions	Today: "Coming soon" toast, no navigation (Z23). After build: open with the active view's records; Sheet View row limit stated (Zoho's Sheet View article: 100 existing rows plus 200 new rows; the list-view FAQ 17 says 999 — record which CRM adopts), edits saved back, record ID column kept	[int]

Group K — Security

ID	Scenario	Expected	Tags
K1	Stored XSS in view name, split name, Kanban name, criteria text value	Rendered as text in picker, tabs, manage pages, dialogs, toasts, audit log, Work Queue	[sec]
K2	IDOR on every verb: GET/PATCH/DELETE/clone/pin another user's private view id 	404/403; no change	[sec] 
K3	Move view to another module: PATCH module_id / use view id under another module path	Refused	[sec]
K4	Enumeration: random UUIDs vs deleted vs private ids	Same 404 body for all	[sec]
K5	Visibility escalation: PATCH own view visibility:"everyone" without permission 	Refused	[sec] 
K6	shared_user_ids with users of another tenant / non-existent	Rejected	[sec]
K7	Criteria that reads hidden data (D21 inference): public view on hidden field 	Per Q13 decision; record	[sec] 
K8	board-summary group_by / aggregate on a hidden field 	Refused; no sums leak	[sec] 
K9	Oversized payloads: 1 MB name, 10 000 columns	413/422, no 500	[sec]
K10	CORS: Origin: https://evil.example preflight on PATCH	Not allowed	[sec]
K11	Tenant isolation: A9 + K6	No cross-tenant read or write	[sec]

ID	Scenario	Expected	Tags
L1	Arabic UI on picker, editor, manage, Kanban, Split	All strings translated; plural strings "{{count}} users" correct in Arabic (two keys missing today — check)	[ui]
L2	RTL layout	Picker, tabs, criteria rows, Kanban column order mirrored; icons rotated	[ui]
L3	Hard-coded English	"Probability", "Layout", "Pipeline" fallbacks — confirm translated in ar	[ui]
L4	Keyboard only	Open picker, create view, edit pattern, move card	[ux]
L5	Screen reader names	Picker "Custom views", "More custom views", "Choose a Kanban view", column buttons	[ux]
L6	Mobile width 390px	Picker and editor usable; no horizontal page scroll	[compat]
L7	Edge + Chrome + Firefox	Core B/C/I flows	[compat]
L8	[perf] list view first page on Leads/Deals	p95 ≤ 1.5 s (PRD target) over 20 loads	[perf]
L9	[perf] Kanban headers + first cards	p95 ≤ 2 s	[perf]
L10	[perf] 15 pinned + 50 views in picker	Picker opens < 500 ms, search responsive	[perf]

E2E journeys

ID	Journey	Steps (chained)	Expected
E2E-1	Sales rep daily list	Create QA CV hot leads (Lead Status any of Interested/Follow Up, created last 7 days, owner = me) → pin → open record → Back → sign out/in	Same view, same results, pinned tab present
E2E-2	Manager shares a working set 	Owner creates view, shares with user → user opens, clones, edits clone → owner edits original	User sees changes in shared view only; clone independent
E2E-3	Pipeline review	Create Deals Kanban by Stage, aggregate Amount → drag QA CV D02 Qualification→Negotiation → open deal → Back	Sums updated, audit entry, same board state
E2E-4	View feeds Work Queue	Create list view → create queue with source = that view → edit view criteria → open queue	Queue reflects edited criteria
E2E-5	Negative recovery	Create view with invalid pattern → fix → save → second tab saves an old copy (conflict) → reload → delete view → restore from Recycle Bin	Each failure has a clear message; final state consistent; no duplicate views
E2E-6	Arabic user	Arabic UI: create view with Arabic name and Arabic text criteria, Kanban in RTL	Works end to end; text and order correct

Group Z — Gaps, run to confirm they are still missing or now built

Z5–Z27 are in NDC-1869. **Z28–Z55 are new in v3** (live Zoho, 2 Oct 2026); each names its item in the missing list Part B and is not in NDC-1869 yet. Every active Z row is expected to fail until the function is built.

Rows Z5–Z27 are in NDC-1869 (22 items, updated 1 Oct 2026); each names its item number. NDC-1869 item order: 1 Z26 · 2 Z27 · 3 Z5 · 4 Z6 · 5 Z7 · 6 Z8 · 7 Z9 · 8 Z10 · 9 Z11 · 10 Z12 · 11 Z19 · 12 Z13 · 13 Z20 · 14 Z21 · 15 Z14 · 16 Z15 · 17 Z16 · 18 Z17 · 19 Z23 · 20 Z24 · 21 Z25 · 22 Z18. Withdrawn in v2 (not gaps, kept here so the numbering stays stable): Z1 (Favourites — current Zoho uses Pin view instead, which CRM has), and Z2, Z3, Z4, Z22 (confirmed not gaps by the user, 1 Oct 2026). Do not run withdrawn rows.

ID	Gap (NDC-1869 item)	How to test	Expected (Zoho / PRD)
Z1	Favourites (removed from NDC-1869) — withdrawn, not a gap	Covered by G5/G6: Pin view puts the view in "Pinned views" at the top of the picker and always shows it as a tab	Current Zoho CRM has removed "Mark as favourite" and replaced it with Pin view (older help pages still describe favourites). CRM has Pin view (up to 15), so CRM matches current Zoho. Removed from NDC-1869 on 1 Oct 2026
Z2	Other users' views for admins (removed from NDC-1869) — withdrawn, not a gap	—	Confirmed not a gap by the user (1 Oct 2026). Removed from NDC-1869 on 1 Oct 2026
Z3	Default view per module (removed from NDC-1869) — withdrawn, not a gap	—	Confirmed not a gap by the user (1 Oct 2026). Last-used view behaviour is tested in B11-B14. Removed from NDC-1869 on 1 Oct 2026
Z4	Recently Viewed standard view (removed from NDC-1869) — withdrawn, not a gap	—	Confirmed not a gap by the user (1 Oct 2026). Standard views are tested in B7. Removed from NDC-1869 on 1 Oct 2026
Z5	Number between / not between (NDC-1869 item 3)	Operator list on Amount	Both present, inclusive bounds documented
Z6	Starting tomorrow / Till yesterday (NDC-1869 item 4)	Date operator list	Both present, open-ended ranges
Z7	Fiscal operators (NDC-1869 item 5)	Date operator list; set Company settings → Fiscal year first	Current / Previous / Next FY and FQ by the tenant fiscal calendar (Zoho live). Live Zoho has no plain calendar-quarter operator, so Last/Current/Next Quarter is only needed if the product owner wants it (NDC-1869 item 5 lists both)
Z8	Share with roles/groups (NDC-1869 item 6)	Share picker kinds	Users, Groups, Roles, Roles and Subordinates (Zoho live; territories only where the org uses them)
Z9	Create-view permission (NDC-1869 item 7)	Profile settings for a "Manage custom views" permission;  user without it	New custom view hidden; API create refused
Z10	Kanban manage permission (NDC-1869 item 8)	Profile setting;  user without it	Create/Edit/Delete Kanban absent; API refused
Z11	Pin (freeze) a column (NDC-1869 item 9)	Column header menu	One column frozen at leading edge (right in RTL)
Z12	Header menu Filter by / Hide (NDC-1869 item 10)	Column header menu	Filter by opens Smart Filters for that field; Hide removes column
Z13	Select all in view (NDC-1869 item 12)	View with > page size records, tick header	"Select all N records in this view", capped, respected by bulk actions
Z14	Sort Kanban cards (NDC-1869 item 15)	Kanban toolbar/settings	Sort cards by chosen field
Z15	Reorder Kanban views + creator on hover (NDC-1869 item 16)	Manage Kanban views	Per-user order; creator/modifier/time on hover

ID	Gap (NDC-1869 item)	How to test	Expected (Zoho / PRD)
Z16	Rollup aggregate (NDC-1869 item 17)	Aggregate list on a module with a rollup field	Rollup selectable, totals correct
Z17	Warn before deleting a queue's source view (NDC-1869 item 18)	G10	Dialog lists dependent queues before delete
Z18	Chart/Timeline/Grid/Map/Canvas views (NDC-1869 item 22)	View-type switcher / More views	Present (non-goal — expected still absent)
Z19	Resize column widths (NDC-1869 item 11)	Drag a column header border in a list view; reload; open as another user 🗝	Width changes and is kept per user per view; other users unchanged (Zoho FAQ 11; NDC-1868 P0-5)
Z20	Row actions menu (NDC-1869 item 13)	Hover a row → "..." menu	Edit, Send Email, Create Task, Add Tags, Change Owner, Convert, Delete, Copy URL, More › Create Call/Meeting/Appointment; filtered by module and permission; Copy URL copies the record link without opening a tab (NDC-1868 P1-6)
Z21	Activity and Notes badges (NDC-1869 item 14)	Lead with open tasks and notes; lead without	Activity badge lists open tasks/meetings/calls/appointments + Create Activity; Notes badge opens Notes panel (view/add/edit/delete/sort), shows on hover when empty; both can be toggled in Manage Columns (NDC-1868 P1-6)
Z22	Converted Leads and Junk Leads standard views (never filed) — withdrawn, not a gap	—	Confirmed not a gap by the user (1 Oct 2026); not to be filed
Z23	"Coming soon" module actions (NDC-1869 item 19)	F12 for Mass Convert, Mass Email, Drafts, Approve, Deduplicate, Add to Campaigns, Create Client Script, Sheet View, Print View	Each opens a working screen scoped to the selection/view (NDC-1868 N3/N4 — non-goals, listed for triage)
Z24	Missing bulk actions (NDC-1869 item 20)	Select records → selection "..." menu	Run Macro, Create Task, Change Owner (many records), Mail Merge present (Zoho live; NDC-1868 N4). "Set reminders" is not in live Zoho and is no longer expected
Z25	System-defined and Website Activity filters (NDC-1869 item 21)	Filter panel → Untouched Records (e.g. 4 weeks), Touched Records, Record Action, Related Records Action, Latest Email Status; Website Activity	Filters apply on top of the view and narrow results (Zoho FAQ 14; NDC-1868 N5). Today "aren't available yet"
Z26	Collapse/expand picker groups (NDC-1869 item 1)	Open the view picker; click each group name (Pinned views, Created by me,	Each group collapses to hide its views and expands to show them again; state is kept while the picker is reopened (record whether it persists

ID	Gap (NDC-1869 item)	How to test	Expected (Zoho / PRD)
		Shared with me, Public views); reopen the picker	across reload). Today the group names are plain labels (user requirement, 1 Oct 2026)
Z27	Manage custom views page grouped by type (NDC-1869 item 2)	Open Manage custom views with own, shared  , public and pinned views; unpin all, reload	Views listed under Created by me, Shared with me, Public views, and Pinned views; the Pinned views group appears only while at least one view is pinned. Today one flat table with "All custom views" / "Public views" tabs (user requirement, 1 Oct 2026; Zoho live groups its Manage page by Created By Me / Public Views)
Z28	Save filter — Save a Filter (missing list B1; not filed)	Apply a field filter on a view → Save filter → clear → click the saved filter → reload	A "Save filter" button appears after a filter is applied; the saved filter is listed, re-applies in one click with the same count, and survives reload (Zoho live, 2 Oct 2026) [func]
Z29	Save filter — Saved Filters Are Personal (missing list B1; not filed)	Save a filter on All Leads as owner → open another view → open All Leads as another user 	Listed only for its creator and only on the view it was saved on (Zoho help: saved filters are user-specific and per view) [func] [sec] 
Z30	Save filter — Manage Saved Filters (missing list B1; not filed)	Save three filters; compare counts; reorder; rename; delete; save until refused	Live count per saved filter equals the grid count; order kept per user; limit stated when reached (Zoho: 5 or 10 per view) [func] [edge]
Z31	Save filter — Saved Filter Name Sweep (missing list B1; not filed)	Save filters named D1, D2, D3, over-long, D8, D11, D19	Empty refused; limit stated; HTML shown as text; duplicate rule recorded [data] [sec]
Z32	Filter By Related Modules — Filter by Related Modules (missing list B2; not filed)	Filter panel → Filter By Related Modules → Tasks with a condition → apply → clear	Section present; filters on top of the view by related records (Zoho live: third section of the filter panel) [func] [int]
Z33	Filter By Related Modules — Related-Module Filter Limits (missing list B2; not filed)	Add related-module filters until refused; combine one with a field filter	Limit stated (Zoho: 3 modules, 5 fields each); combined filters are ANDed [edge]
Z34	Lock this View — Lock Toggle (missing list B3; not filed)	New custom view → Only me vs Everyone → turn Lock this View on → Save → reopen → GET the view	Toggle shown only for shared views with "Restrict any changes by users with whom the view is shared."; state saved and returned by the API (Zoho live) [func]
Z35	Lock this View — Locked View Blocks Changes (missing list B3; not filed)	As a second user with Manage Shared Views, try Edit, Save to view and Delete on a locked view; then edit as creator and as admin 	Others cannot change name, criteria, columns or sharing or delete; creator and administrators can (Zoho API: "only Admins and creators can modify it") [sec] 

ID	Gap (NDC-1869 item)	How to test	Expected (Zoho / PRD)
Z36	Lock this View — Locked View in the API (missing list B3; not filed)	As a non-creator: PATCH name, PATCH locked=false, DELETE on a locked view 	All refused; view unchanged (name, lock, row_version) [sec] 
Z37	Lock this View — Unlock and Clone (missing list B3; not filed)	Clone a locked view as another user; unlock as creator; edit as the other user; set to Only me 	Clone is private and unlocked; unlock restores editing; lock is cleared when the view is no longer shared [func] [edge] 
Z38	Owner criteria by role — Owner Belongs to Role (missing list B4; not filed)	Lead Owner operator list → belongs to Role → pick a role → Save 	Operators "belongs to Role" and "does not belong to Role" offered; results are the records owned by users in that role (Zoho live) [func] 
Z39	Owner criteria by role — Owner Role Edge Cases (missing list B4; not filed)	does not belong to Role; role with no users; user moved to another role; unknown role id via API 	Complement set returned; empty role = empty view; results follow the current role; unknown id refused [edge] [sec] 
Z40	Date criteria Previous / Next N units — Previous and Next N Units (missing list B5; not filed)	Date operator list → Previous / Next → number + unit (days, weeks, months, years) → Save each	Operators offered with the four units; each returns only records inside the range (Zoho live) [func]
Z41	Date criteria Previous / Next N units — Previous and Next Boundaries (missing list B5; not filed)	Previous with D1, 0, 1, max, max+1, -1, 1.5, text, Arabic-Indic digits — UI and API	Invalid numbers refused in UI and API; limits stated; no 500 [data] [sec]
Z42	Date criteria Previous / Next N units — Previous and Next at Period Edges (missing list B5; not filed)	Previous 1 day / week / month with records at 23:50 and 00:10 Cairo around each edge	Edges follow the tenant timezone; the rule for the current period is consistent and stated (Zoho API excludes it) [edge]
Z43	Text operators on pick lists — Text Operators on Pick Lists (missing list B6; not filed)	Lead Status operator list → contains / doesn't contain / starts with / ends with	The four text operators are offered on pick lists and match on the option label (Zoho live) [func]
Z44	Text operators on pick lists — Pick List Text Operator Sweep (missing list B6; not filed)	contains with D7, D6, D8, D12, D11; Arabic UI; removed option	Same matching rule as text fields; input treated as text; same results in Arabic; no crash [data] [sec]
Z45	Wrap Text view mode — Wrap Text View Mode (missing list B7; not filed)	Column settings menu → View Mode → Wrap Text / clipped with a 255-character value → reload	View Mode offered; wrap shows the whole value, clip shows one line; choice kept (Zoho live: "View Mode — Wrap Text") [func] [ui]
Z46	Wrap Text view mode — Wrap Text Is Personal (missing list B7; not filed)	Change wrap mode on a shared view as owner; open as	Per user; shared definition unchanged; correct in RTL [func] [ui] 

ID	Gap (NDC-1869 item)	How to test	Expected (Zoho / PRD)
		another user; check row_version; Arabic text 🔒	
Z47	Reset Column Size — Reset Column Size (missing list B8; not filed)	Resize two columns → column settings menu → Reset Column Size → reload	All widths back to default and kept; order, visibility and sort untouched (Zoho live) [func]
Z48	A–Z letter filter — A–Z Letter Filter (missing list B9; not filed)	Name column "All" dropdown → pick a letter → pick a letter with no records → All	Options All, A–Z; only names starting with the letter; count updates; All resets (Zoho live) [func]
Z49	A–Z letter filter — Letter Filter Combined (missing list B9; not filed)	Letter + view criteria + paging + sort + field filter; change view; Arabic UI; names starting with a digit	Narrows on top of the view; kept through paging and sort; cleared on view change; rule for Arabic and digits recorded [edge] [ui]
Z50	Delete several views at once — Delete Several Views (missing list B10; not filed)	Manage custom views → tick two views → Delete → Cancel → Delete → confirm	Checkbox per custom view; Delete appears; confirmation counts the views; only the ticked views are deleted (Zoho live) [func]
Z51	Delete several views at once — Bulk Delete Protections (missing list B10; not filed)	Try to tick a system view and a view shared by someone else; bulk delete a view that a queue uses 🔒	System and non-deletable views cannot be selected; dependent queues are listed before deletion [sec] [int] 🔒
Z52	Delete several views at once — Bulk Delete in the API (missing list B10; not filed)	Bulk delete with a valid id, a system view id and an unknown id; send twice in parallel	Per-id result; system and unknown ids refused; no partial surprise; no 500 [sec] [edge]
Z53	More selection and module actions — More Selection Actions (missing list B11; not filed)	Select two records → selection "..." menu → Cadences, Print Mailing Labels, Print Using Canvas	Each is offered and opens a screen scoped to the selection (Zoho live; NDC-1868 N4 non-goal) [int]
Z54	More selection and module actions — Export Selected Records (missing list B11; not filed)	Tick three records → Export Selected Records → open the file → clear the selection	Only the selected records are exported; formula-like values are neutralised; needs a selection [int] [sec]
Z55	More selection and module actions — Assignment Rules and Mass Transfer (missing list B11; not filed)	Module actions menu → Assignment Rules; Mass Transfer up to the preview	Both offered; Mass Transfer previews the records before changing owner (Zoho live; NDC-1868 N4 non-goal) [int]

PRD = NDC-1868 (custom-views-spec.md). CV = existing xlsx ids.

PRD requirement	Scenarios	Existing CV ids
P0-1 Picker, categories, standard views	B1–B16, A12, G5, G6, Z26, Z27	CV-001–CV-009
P0-2 Create/edit list view	C1–C23, A2, G5, Z9	CV-010–CV-025
P0-3 Operators	D1–D19, Z5–Z7	CV-026–CV-054
P0-4 Pattern	D20–D24	CV-055–CV-067
P0-5 Columns and in-view controls	C9–C12, F1–F8, F16, F17, E11, Z11, Z12, Z19	CV-068–CV-078
P0-6 Sharing and access	A2–A7, C14, C15, E10, G13, K2, K5–K7, Z8	CV-079–CV-090
P0-7 Clone and delete	G1–G13, Z17	CV-091–CV-099
P0-8 Selection and bulk scope	F9–F13, I16, Z13	CV-100–CV-106
P0-9 Kanban core	H2–H4, H6, H10, I1–I20, Z14	CV-107–CV-130
P0-10 Kanban management	H9, H12, H13, A8, Z10, Z15	CV-131–CV-137
P1-1 Related-module criteria	E1–E7	CV-138–CV-142
P1-2 Subform criteria	Inconclusive (§1.4)	CV-143–CV-145
P1-3 Lookup criteria	E8	CV-146–CV-147
P1-4 Fiscal operators	Z7	CV-148–CV-149
P1-5 Last Activity Time	D25	CV-150–CV-151
P1-6 Row actions and badges	F14, F18, I15, Z20, Z21	CV-152–CV-156
N3 Sheet and Print views (non-goal)	F12, J5, Z23	— (new)
N4 Bulk operations (non-goal, scoping only)	F11, F12, Z23, Z24	— (new)
N5 Smart Filters coexist with views (non-goal)	F19, Z25	— (new)
P1-7 Kanban refinements	H6, H8, H14, Z16	CV-157–CV-160
NFR i18n/RTL	L1–L3, D24, E2E-6	CV-161–CV-163

PRD requirement	Scenarios	Existing CV ids
NFR a11y + keyboard drag	L4, L5, I13	CV-164–CV-165
NFR audit	C24	CV-166
NFR performance	L8–L10	CV-167–CV-168
Instrumentation (§10)	not testable from UI — needs analytics access; listed as a hole	CV-169–CV-174
Field lifecycle (§8)	E9	CV-175–CV-176
Count = rows	F8, I1	CV-177
Request tampering	K2–K4	CV-178
Empty view	F15	CV-179
Beyond PRD (built): Split view, version conflict, Recycle Bin, last N hours	J1–J4, C18, G8, D11	— (new)
Beyond PRD (new gaps, missing list Part B): saved filters and related-module filters	Z28–Z33	— (new)
Beyond PRD (new gaps): Lock this View	Z34–Z37	— (new)
Beyond PRD (new gaps): criteria — owner by role, Previous / Next N units, pick list text operators	Z38–Z44	— (new)
Beyond PRD (new gaps): grid — Wrap Text, Reset Column Size, A–Z letter filter	Z45–Z49	— (new)
Beyond PRD (new gaps): delete several views at once	Z50–Z52	— (new)
Beyond PRD (new gaps): more selection and module actions (N4 non-goal)	Z53–Z55	— (new)

Holes: instrumentation events (no way to observe from QA) and P1-2 subform (no subform module) — both listed in NDC-1869 "Could not be checked" or here. Requirement-by-requirement coverage is in `custom-views-requirements-v3.md` .

Reporting

- Folder: `Test Cases/CRM/Results/Custom Views - execution <date>/` with `report.md` , per-scenario screenshot, API request/response (redacted).
- Redaction: never write tokens, cookies or passwords; write `Bearer <len=1263>` . Mask emails of other users.
- Each Fail: steps, expected, actual, evidence, and whether it maps to an existing NDC item (NDC-1836, NDC-1317, NDC-1869 Zn) before filing anything new — run the false-positive checklist (feature flag, permission, layout scope, cached SPA, stale token).
- Coverage gaps restated at the end: 🚧 scenarios and subform scenarios Inconclusive while §1.4 blockers stand.
- Revert proof: after cleanup, repeat every §1.3 GET and diff against the baseline. Only expected differences allowed (Recycle Bin entries, if not purged, listed by id). Attach the diff.